

order to limit the coverage area, which also increases the possibility of frequency re-use. Moreover, the radiation level also increases in the proximity of antenna. So, it is advised to decrease VRP downtilt in order to reduce the radiation level. The downtilt of the main beam can be achieved by either mechanically or electrically-operated motor mechanism.

4. Increase in antenna gain. Gain of an antenna is a key performance figure that combines the antenna's directivity and electrical efficiency ($G = \eta D(\theta, \phi)$). It specifies how well the antenna converts input power into electromagnetic waves headed in a specified direction and to limit the radiation in other directions. So, it is possible to limit the radiation level in the area accessible to the people with the control of the directivity (ultimately gain) of the antenna. Again, the directivity of the antenna is related to horizontal radiation pattern (HRP) and vertical radiation pattern (VRP). In order to protect people against radiation, HRP of the antenna can be altered but it will always affect the coverage area.

On the other hand, VRP determines radiation as a function of distance to the antenna. So, antenna gain (or more precisely VRP) can be utilised (increased gain/sharp main lobe) to limit the radiation level in close proximity of the antenna without affecting the coverage area. This can be achieved by feeding a high-gain antenna with a low-power transmitter, which will result in the same amount of equivalent radiated power (ERP). So, it is recommended to use a low-power transmitter with a high-gain antenna (narrower VRP beam width) for attaining a low-level of radiation.

5. Changes in VRP. VRP is a function of elevation angle (i.e., in a vertical plane) and represents the distribution of energy in a vertical plane. It gives an impression of distribution of energy depending on the distance between a transmitting antenna and an observation point. Vertical radiation



Fig. 5: Signage for placement at mobile phone tower site

pattern is responsible for radiation to the area in the proximity of the antenna. So, changes can be done in VRP by introducing changes in the feeding arrangement (changing cable length). So, an optimisation of VRP will result in reduction in radiation level.

6. Changes in HRP. HRP is a function of azimuth angle, representing distribution of energy in a horizontal plane. It is possible to reduce the radiation level by using an antenna with a narrower horizontal beam. A narrower horizontal beam (means higher antenna gain) accompanied with lower transmitter power will result in lower radiation level without the loss of coverage area.

7. Multiple methods applied simultaneously. All methods described above are independent and can be applied either individually or in combination to achieve the required decrease in radiation level.

Placement of signage

It is the responsibility of the wireless service provider to ensure provision of proper signage, warning entry of general public to the exclusion zones. Samples of different signboard are shown in Fig. 5. The danger signboard should be placed on the tower structure. Warning signboard should be placed at the entry point of the exclusion zone and the caution signboard should be placed at the entrance of the base station compound in case of ground base tower (GBT) or at the entry point of the

roof of the building in case of roof-top tower (RTT).

Conclusion

Operators providing wireless communication should seriously consider this study and ITU-T recommendations in order to keep the operation of base-station transceivers in compliance with regulations concerning environmental protection against non-ionising radiation. Further, it is important to note that the present threshold limits prescribed by ICNIRP are considered to be rather too generous. Hence, there is a need to review and remedy the situation and not wait until it becomes the subject matter of a public-interest petition in the light of possible environmental adverse effects.

In order to minimise possible health hazards, some recommendations are: minimisation of mobile phone usage, limitation of use by at-risk population (such as children), adoption of mobile phones and microcells with as low as reasonably-practicable levels of radiation, wider use of hands-free and earphone technologies such as Bluetooth headsets, headsets with ferrite beads and headsets with air tube, adoption of maximal standards of exposure and greater distances of base-station antennae from human habitations, and so forth. ●

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